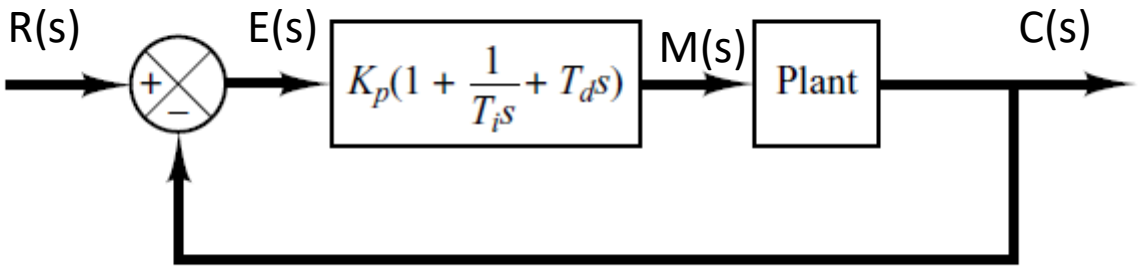
Control Design Using Root Locus

Objectives

- To understand PID controller and its mathematical expression
- To recognize and understand PD and PI controllers
- To understand phase lead (PD approx.) and phase lag compensators (PI approx.)
- To understand lead-lag phase controllers.
- Design phase lead, lag and lead-lag controllers using root-locus techniques

PID controller

- The PID controller is depicted in the following block diagram



- That is,

$$G_c(s) = \frac{M(s)}{E(s)} = K_p(1 + \frac{1}{T_i s} + T_d s)$$

PID Controller

- Notice that the output is the sum of three parts: proportional, integral and derivative parts:

$$K_p e(t)$$

$$\frac{K_p}{T_i} \int e(t)dt$$

$$K_p T_d \frac{de}{dt}$$

PID Controller

- Notice also

$$G_c(s) = K_p\left(1 + \frac{1}{sT_i} + T_d s\right) = \frac{K_p}{T_i}\left(\frac{s^2 T_d T_i + sT_i + 1}{s}\right)$$

- Therefore, the PID transfer function is not realizable because the numerator polynomial order is bigger than the order of the denominator (Kp/Ti is the static gain relayed to the steady state error)
- A real PID would be then:

$$G_c(s) = \frac{K_p}{T_i}\left(\frac{s^2 T_d T_i + sT_i + 1}{s(T_f s + 1)}\right)$$

where T_f should be as small as possible.

PD controller

- PD controller is a proportional-derivative controller

$$G_c(s) = K_c(sT_d + 1)$$

- We can notice that the PD is not physically realizable for the same reason a PID is not physically realizable.
- A real PD would include a pole as follows

$$G_c(s) = K_c\left(\frac{1 + sT_1}{1 + sT_2}\right) = K_c\frac{T_1}{T_2}\left(\frac{s + \frac{1}{T_1}}{s + \frac{1}{T_2}}\right) \quad T_1 > T_2;\ \ \alpha = \frac{T_1}{T_2}$$

- The controller of the previous bullet (PD approximation) is also called a phase lead compensator/controller. For the phase lead controller to approximate closely a PD, $\alpha \gg 1$ or $T_1 \gg T_2$

PI Controller

- PI controller is a proportional-integral controller as follows

$$G_c(s) = K_p\left(1 + \frac{1}{sT_i}\right) = \frac{K_p}{T_i}\left(\frac{sT_i+1}{s}\right)$$

- PI is physically realizable. (Kp/Ti is the static gain related to steady state error)
- An approximation of the PI is the phase lag compensator/controller

$$G_c(s) = K_c\left(\frac{1+sT_2}{1+sT_1}\right) = K_c\frac{T_2}{T_1}\left(\frac{s+\frac{1}{T_2}}{s+\frac{1}{T_1}}\right);\ T_1 > T_2;\ \ \beta = \frac{T_1}{T_2}$$

- For the phase lag controller to be a good approximation of a PI, β has to be large (but not too large either): $\beta \gg 1$ or $T_1 \gg T_2$

PID Approximation

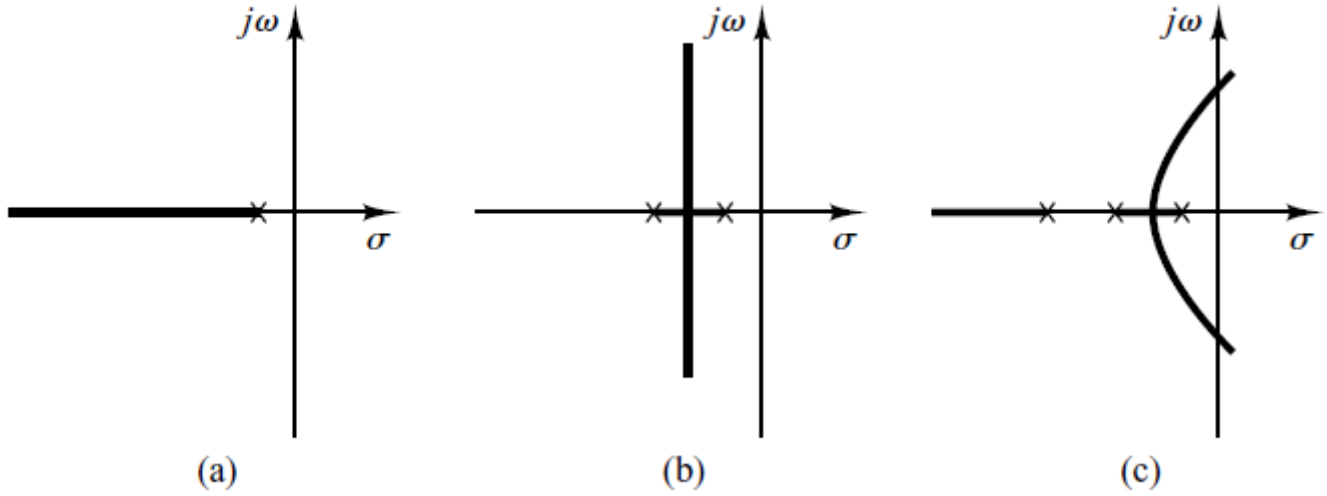
- We could think that a phase lead-lag controller/ compensator is the approximation of a PID:

$$G_c(s) = K_c \underbrace{\left(\frac{1+\alpha T s}{1+Ts}\right)}_{\text{lead}} \underbrace{\left(\frac{1+T's}{1+\beta T'}\right)}_{\text{lag}}$$

- It's the product of the two compensators. Remember that $\alpha > 1$ and $\beta > 1$.

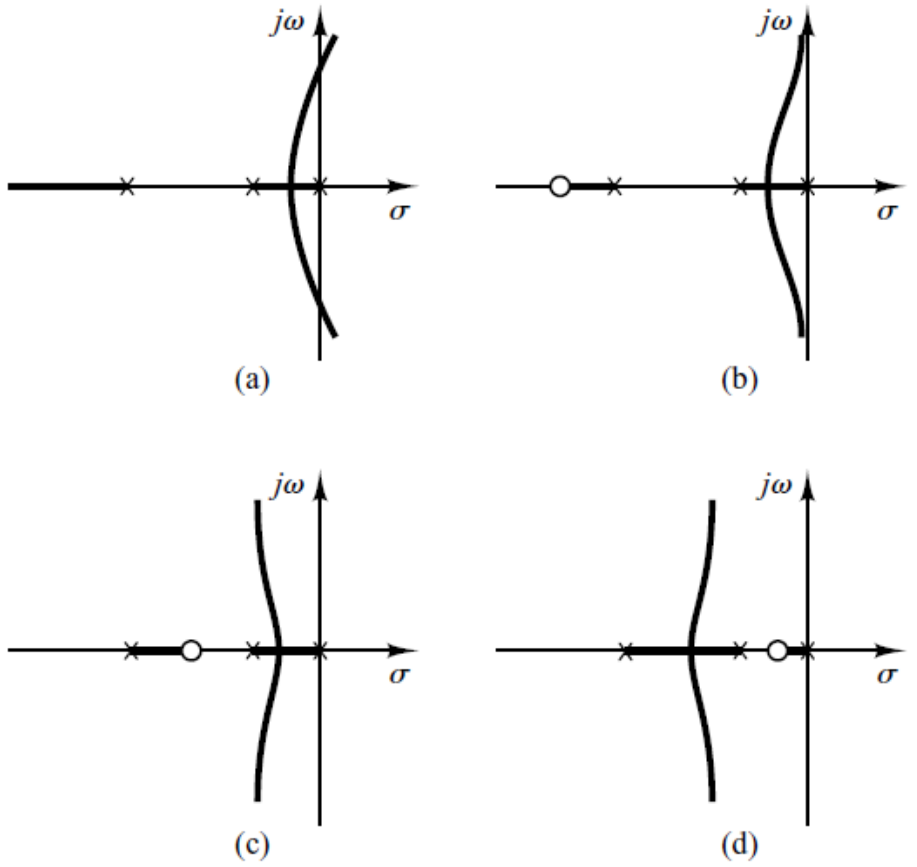
Effects of Adding Poles

- The addition of a pole to the open-loop transfer function has the effect of pulling the root locus to the right, tending to lower the system's relative stability and to slow down the settling of the response.



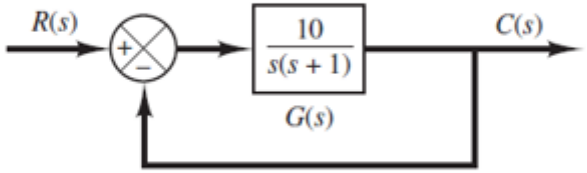
Effects of Adding Zeros

- The addition of a zero to the open-loop transfer function has the effect of pulling the root locus tp the left, tending to make the system more stable and to speed up the settling time.

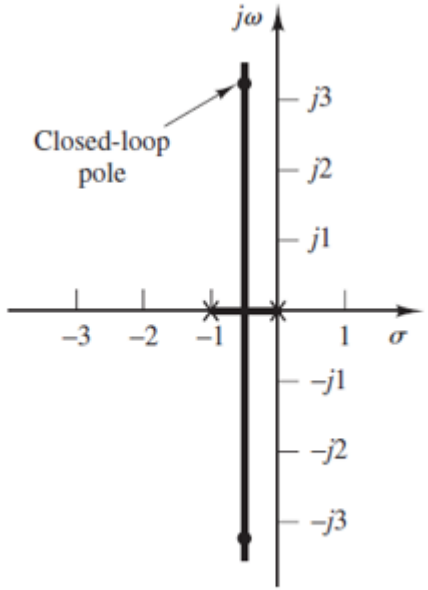


Designing Controllers Using Root-Locus (Problems 1/2)

- Design a phase-lead controller for the following system



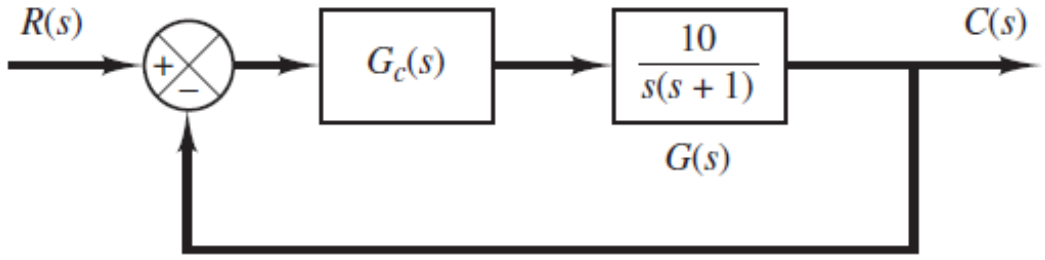
(a)

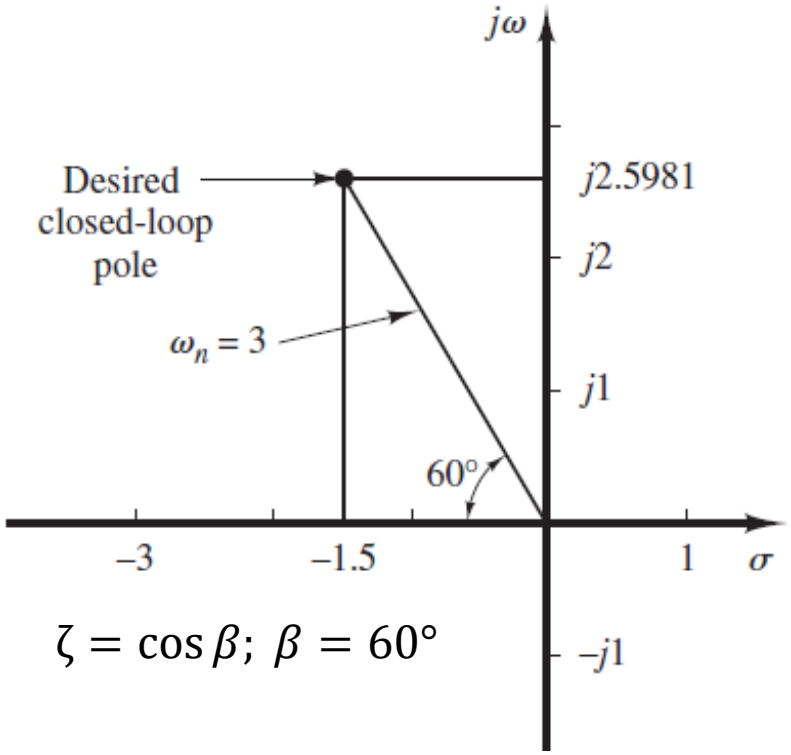


(b)

Designing Controllers Using Root-Locus (Example 1/2)

- It is desired to design the lead compensator such that the dominant closed-loop poles have the damping ratio $\zeta = 0.5$ and the undamped natural frequency $\omega_n = 3\ \mathrm{rad/s}$. It is also desirable that the velocity constant (Kv) be as high as possible.

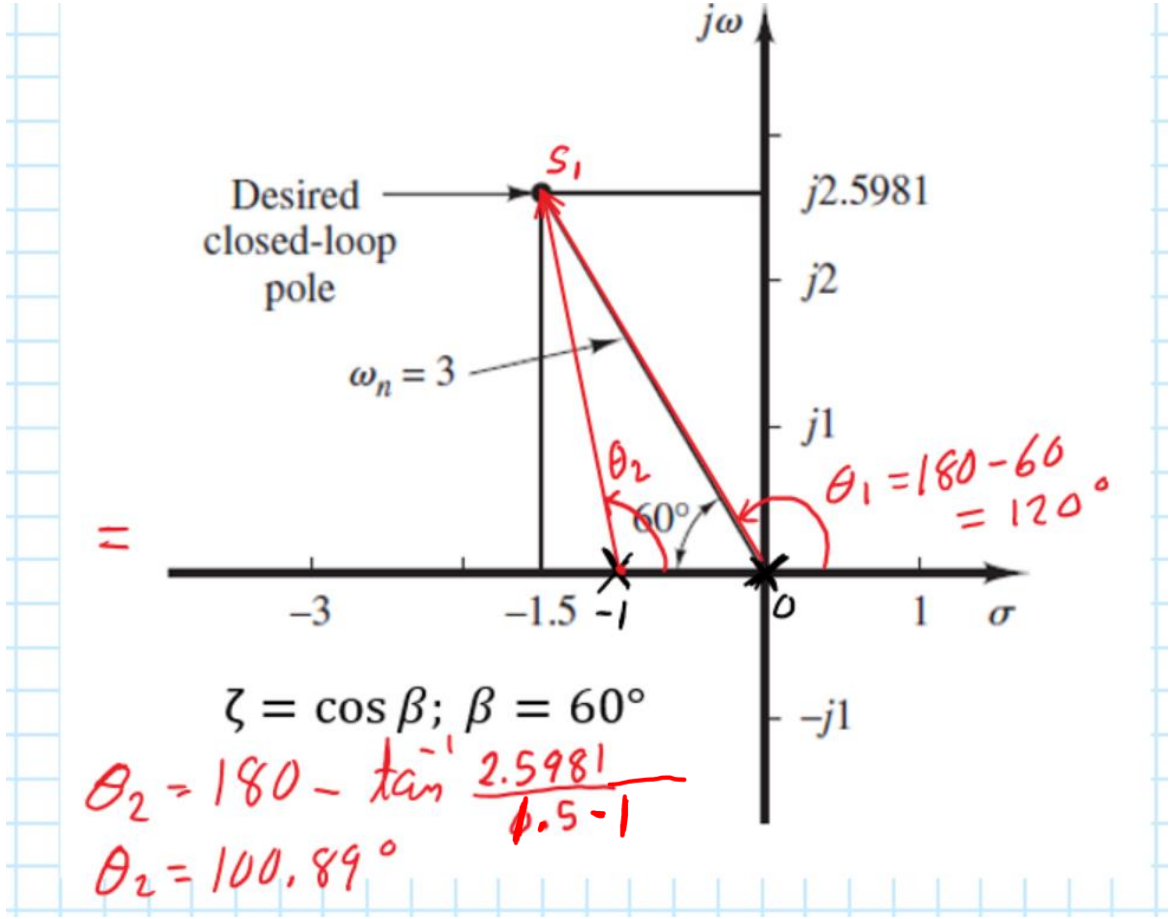




Control Design Principles Using Root Locus

- In order for the desired poles $s_{1,2} = -1.5 \pm j2.5981$ to be poles of the closed loop poles (controlled system), they need to satisfy the angle condition. Let's see if they do it (actually, it's enough to take one of them, typically the one with positive imaginary part, $s_1 = -1.5 + j2.5981$).
- Following the root locus procedure, we draw the vectors from the open loop poles and zeros (no zeros in this example) to the desired closed loop pole location, and then verify the angular contribution.

Control Design Principles Using Root Locus



Control Design Principles Using Root Locus

- We verify then, that the angular contribution is, $-\theta_1 - \theta_2 = -120° - 100.89° = -220.89°$
- We conclude then that s_1 is not in the root locus (nor is s_2) because the angular contribution is not -180 or 180 (or any odd multiple of these numbers).
- The angle deficiency is what we need to add for that point s_1 to become part of the root locus. Because -180 is closer to -220.89 than +180, we do:

$$\measuredangle def - 220.89° = -180° \Rightarrow \measuredangle def = \gamma = 40.89°$$

Control Design Principles Using Root Locus

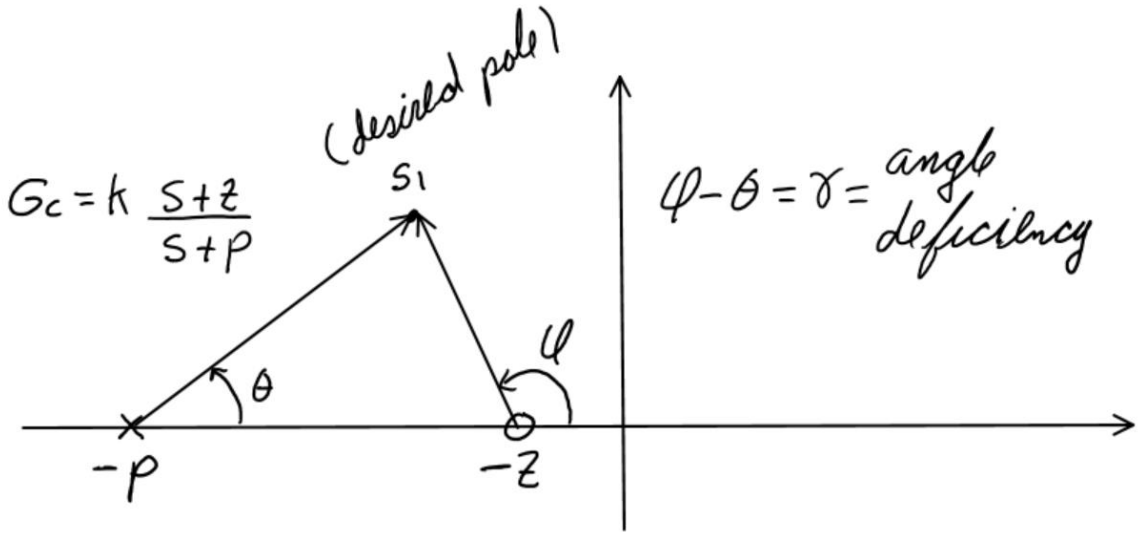
- What would happen if we take +180?

$$\measuredangle def - 220.89° = 180° \Rightarrow \measuredangle def = \gamma = 400.89°$$

- 400.89 is a big number; it's bigger than 360. Because adding 360 to any angle gives us the same angle position, the angle deficiency must be less than 360. The right answer is $\measuredangle def = \gamma = 400.89° - 360° = 40.89°$ (400.89 module 360)
- In other words, the controller has to supply γ =+40.89° for s_1 to become a pole of the closed loop system.
- Only a phase lead controller can add the positive angle of 40.89. A phase lag controller introduces a negative angular contribution. See next figure.

Control Design Principles Using Root Locus

- A phase lead controller has a zero and a pole in the LHP (left half plane) of the s-plane. The zero must be closer to the origin as follows



The pole-zero arrangement of the phase lead controller must be such that its angular contribution is exactly the angle deficiency.

Control Design Principles Using Root Locus

- Notice also that there are many possible solutions to this problem.
- For example, you can fix the zero of the phase lead controller anywhere on the right of s_1 (exclude the origin), and then find the pole to make sure the compensator angular contribution matches the angle deficiency (Problem 2, we will illustrate this in class).
- You can also fix the zero of the phase lead controller on the left of s_1, but the location cannot be too far off from s_1, why?
- However, there's an optimal method to find the zero/pole location of the phase lead controller that guarantees the maximum possible position gain. This method is explained in another set of slides (*Optimal Phase Lead Controller.pdf*). We will illustrate this in class.

Phase-Lead Design – Non-Optimal Method

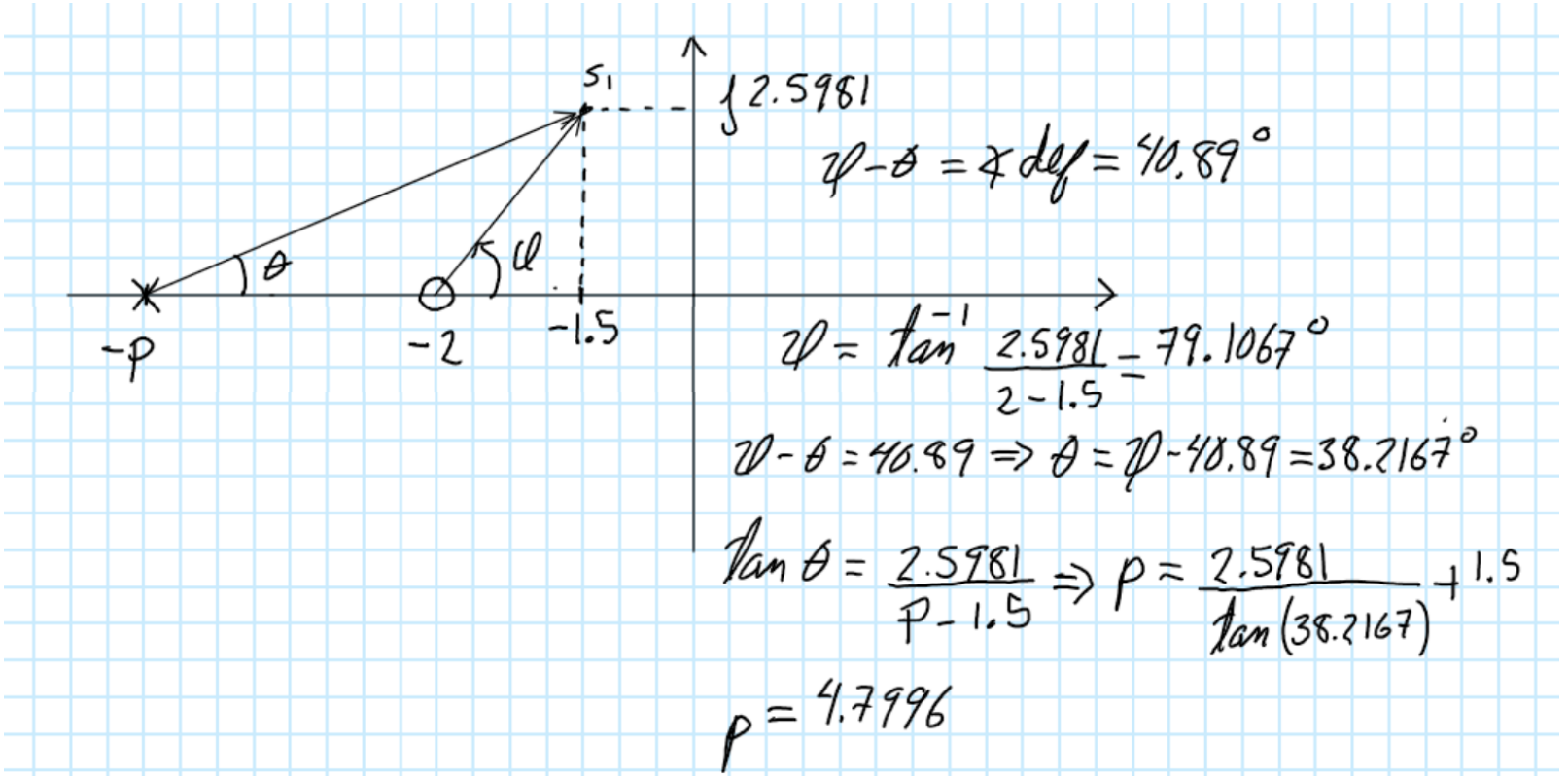
- Step 1. Converts the information about the desired closed loop poles into the real location of such poles in the s-plane. We did that in slide 13 knowing that $s_{1,2} = -\zeta\omega_n \pm j\omega_n\sqrt{1-\zeta^2} = -1.5 \pm \mathrm{j}2.5981$.
- Step 2. Draw the pole-zero diagram of the open loop transfer function together with one of the poles found in Step 1 (usually the one with positive imaginary part) and calculate the angle deficiency. We did this in slides 14 and 15, where $\angle\, def = 40.89°$.
- Step 3. If the angle deficiency is bigger than zero, use a phase lead controller. Not “so arbitrarily” (ask, why not “so arbitrarily”?) select the zero of the phase lead. In this case, we choose a zero at $s = -2$. Hence, the transfer function of our controller is so far like this:

$$G_c(s) = K\frac{s+2}{s+p}$$

Phase-Lead Design – Non-Optimal Method

- Step 4. Find the value of p to make the angle deficiency zero with the inclusion of the controller (see slide 17). In this case we have, the controller transfer function:

$$G_c(s) = K\frac{s+2}{s+4.8}$$



Phase-Lead Design – Non-Optimal Method

- Step 5. While the angle condition is satisfied by the location of the zero and pole of the phase-lead controller, the magnitude condition is satisfied with the calculation of K as follows:

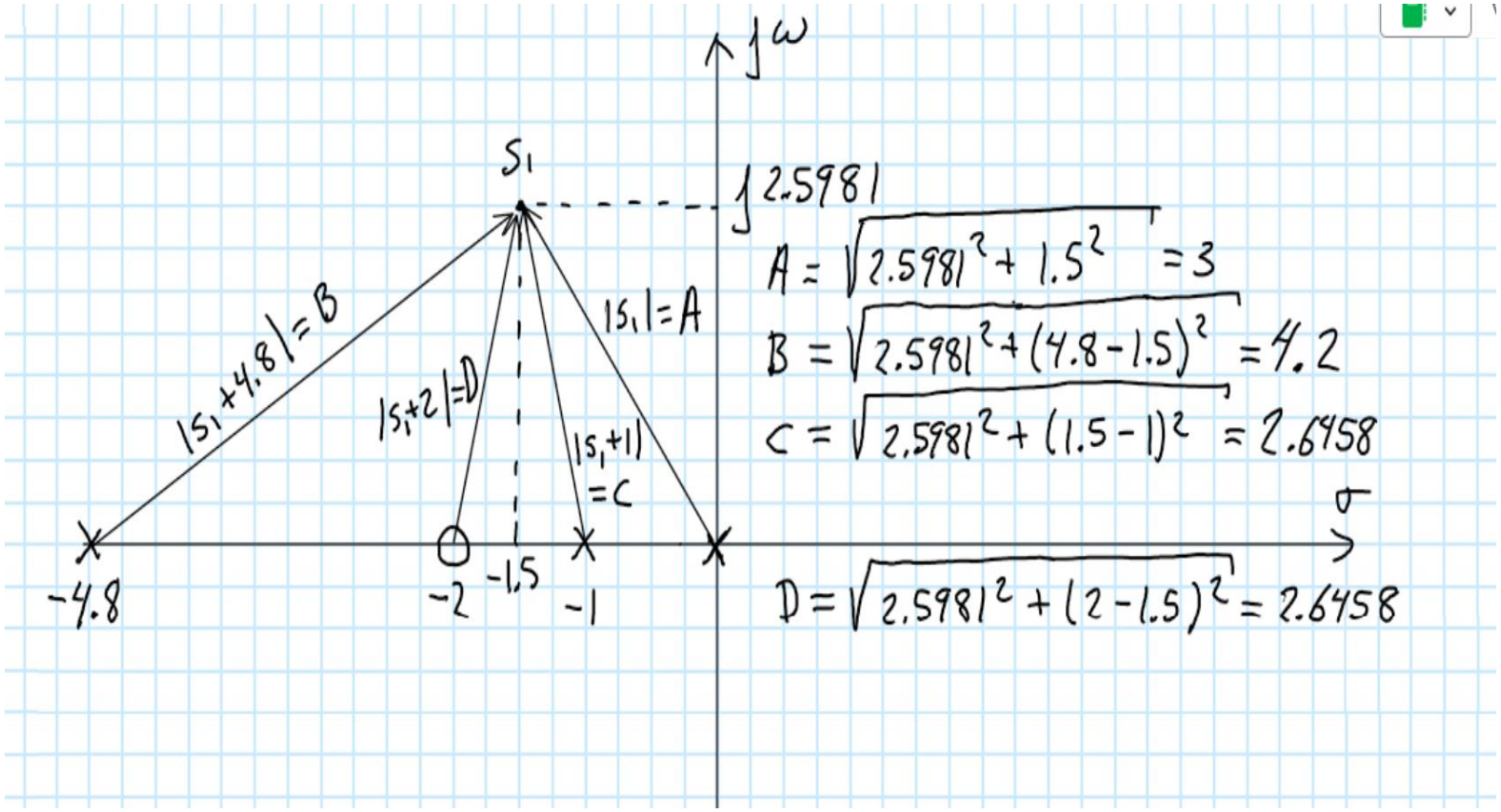
$$|G_c(s)G(s)H(s)|_{s=s_1} = \left| K \frac{10(s+2)}{s(s+4.8)(s+1)} \right|_{s=s_1=-1.5+j2.5981} = 1$$

$$K = \frac{|s_1||s_1+4.8||s_1+1|}{10|s_1+2|} = \frac{ABC}{10D}$$

Phase-Lead Design – Non-Optimal Method

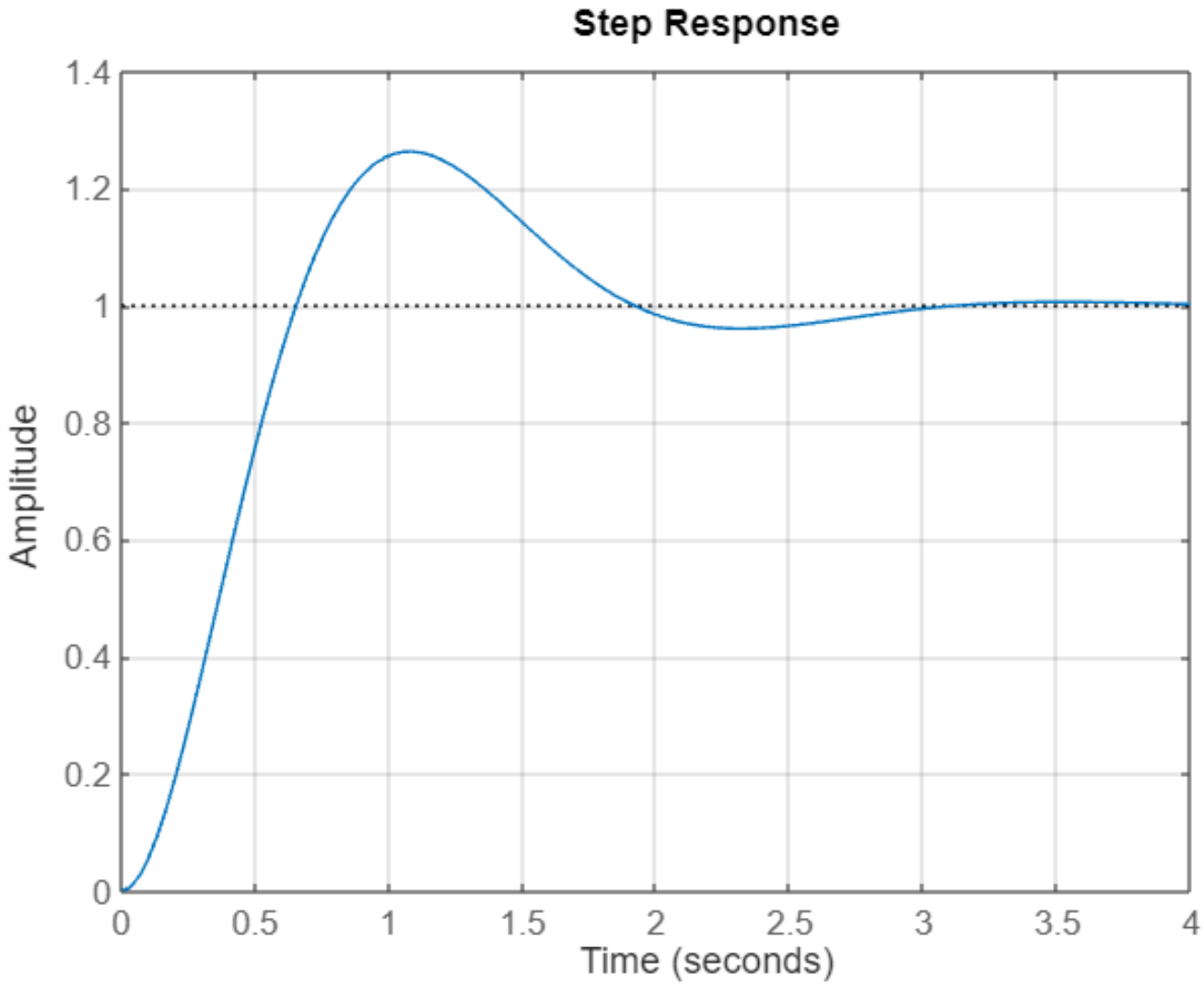
- $K = \frac{3 \times 4.2 \times 2.6458}{10 \times 2.6458} = 1.26$, so the final controller is

$$G_c(s) = 1.26 \frac{s+2}{s+4.8}$$



Step Response with Controller Designed Previously

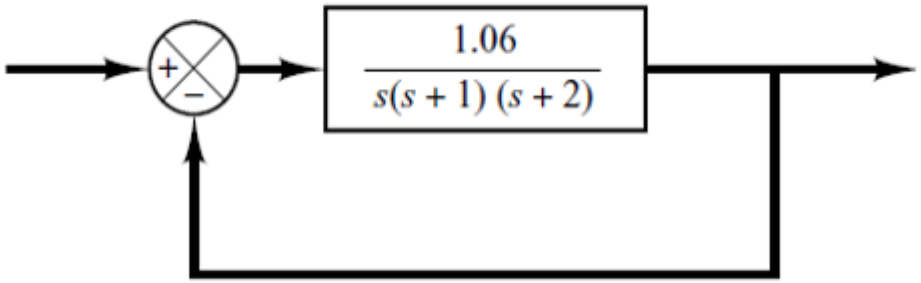
- See the use of MATLAB for this problem (*phaseLeadDesignNonOpt.pdf*)

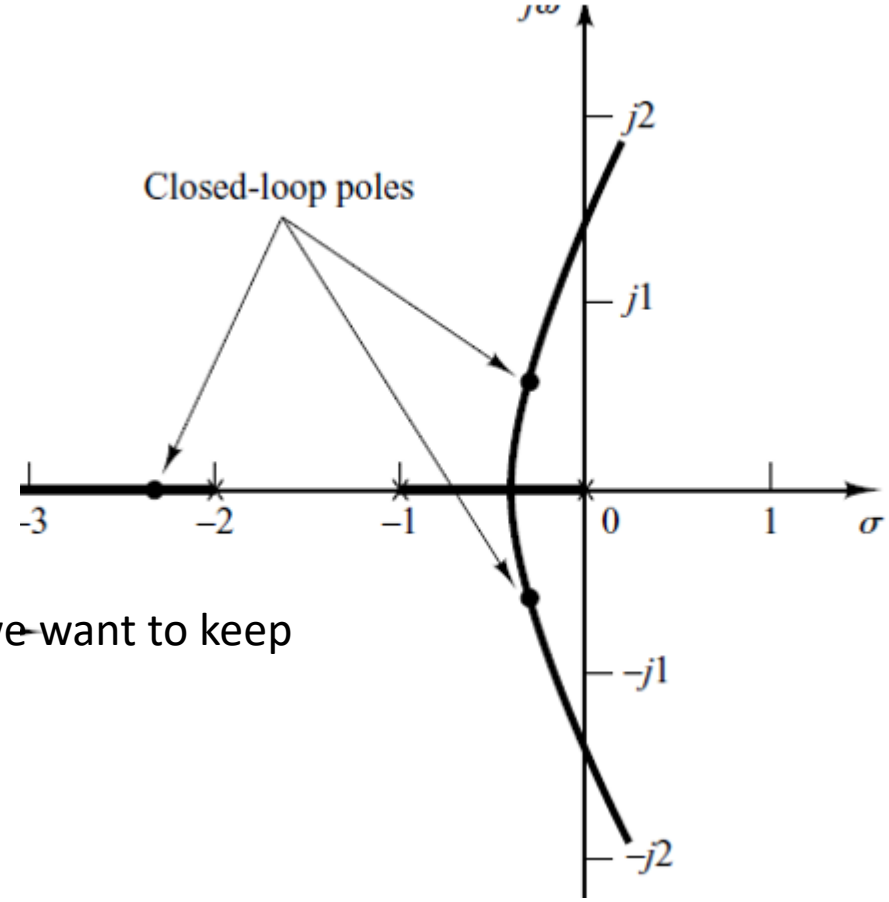


Problem 3 – Lag Compensator Design

- Using the root locus to design phase lag controllers is limited. It only can be used when the closed loop poles are good, and we don't want to change them; however, the gain of the system is not good, and we need to increase it to reduce the steady state error.
- For example, consider the system in the following slide. Its dominant poles are $s_{1,2} = -0.3307 \pm j0.5864$, which are good (why?), but it's Kv = 0.53 is deemed to be too low.
- The idea is then to design a controller that increases the gain Kv 10 times and approximately keeps the current dominant closed loop poles (see handout for solution).

Problem 3 – Lag Compensator Design

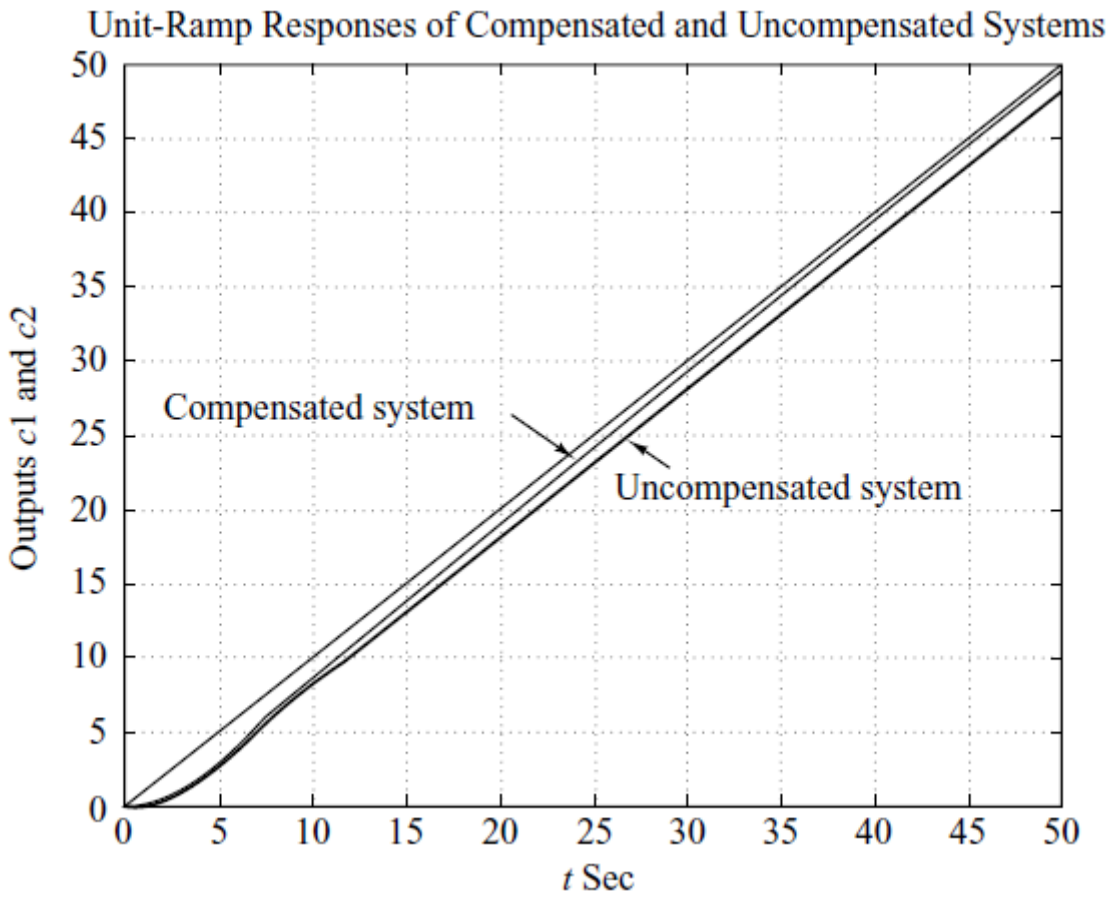




This system has good closed loop poles we want to keep

Example 3 – Lag Compensator Design

- After the solution



Lead- Lag Compensators

- The design of lead-lag compensators involves the techniques we have indicated for the lead and lag compensators.
- You will require this type of controller when both your controller must add a positive angular contribution to guarantee the location of the dominant poles and needs to adjust the gain to satisfy a required steady state error.
- You will design the phase lead to guarantee the pole location, and then a lag controller (that takes into account the lead controller) to guarantee the right value of the gain (see example to be solved in class and in the handout).